

Quarkus 3.15

Java 21

Nuxt 3

TypeScript 5

PostgreSQL 16

Docker

AES-256 / RSA-2048

Microservice Ready

Native Binary

Multi-Module Maven

Ringkasan Proyek

enterprise-libs adalah monorepo framework enterprise yang menyediakan modul-modul siap pakai untuk backend berbasis **Quarkus/Java** dan frontend berbasis **Nuxt 3/TypeScript**. Framework ini dirancang untuk mempercepat pengembangan aplikasi enterprise dengan mengikuti arsitektur microservice, containerization Docker, enkripsi data tingkat enterprise, dan sistem tema dinamis berbasis design tokens. Setiap modul berdiri sendiri, memiliki Dockerfile, docker-compose, dan README yang lengkap.

REPOSITORY

JackD-Creator/Java-Devx

github.com · branch: master

BACKEND STACK

Quarkus 3.15.1 + Java 21

Maven multi-module · Flyway · Panache

FRONTEND STACK

Nuxt 3 + Vue 3 + TypeScript

npm workspaces · @nuxt/module-builder

DATABASE

PostgreSQL 16

Hibernate ORM Panache · Flyway migrations

CONTAINER

Docker (Rancher Desktop)

docker-compose per modul · WSL 2

LISENSI & STATUS

MIT · v1.0.0-SNAPSHOT

Production-ready · Mei 2026

Modul Backend (3 Modul)

quarkus-db-adapter

Abstraksi database enterprise untuk Quarkus

- BaseEntity — id SEQUENCE/50, createdAt, updatedAt, createdBy, updatedBy
- PageResult<T> — generic paginasi: data, total, page, size, totalPages
- BaseRepository<E> — findAllPaginated, findByQueryPaginated, saveOrUpdate
- DatabaseHealthCheck — @Liveness @Readiness via JDBC isValid(2)

quarkus-encryption

Library enkripsi enterprise multi-algoritma

- AES-256-GCM — 12-byte random IV, output Base64(IV+ciphertext)
- RSA-2048 OAEP — generateKeyPair, loadPublicKey/loadPrivateKey dari Base64
- BCrypt password hash (cost=12) + SHA-256/512 checksum
- HMAC-SHA256 + safeEquals() constant-time comparison
- REST endpoint opsional (toggle via application.properties)

quarkus-theme-api

REST API manajemen tema dinamis berbasis design tokens

- Theme model — PanacheEntity: name, slug, active, mode, tokensJson (JSONB), tenantId
- 7 REST endpoint: GET /themes, GET /active, GET /{id}, POST, PUT, PUT /activate, DELETE
- Multi-stage Dockerfile build dari monorepo root context
- ThemeService — CRUD + activate() dengan deactivate-all-others logic
- Flyway migration — seed tema *default* (light) & *dark* lengkap
- docker-compose dengan PostgreSQL 16 healthcheck

nuxt-theme-lib

Nuxt module tema dinamis dari API

- Fetch tema aktif saat boot, SSR-safe via useState
- applyToDocument() — tulis CSS custom properties ke <html>
- useTheme() — fetchActiveTheme, fetchAllThemes, setThemeLocally
- Tipe: ThemeTokens, ThemeColors, ThemeTypography, ThemeSpacing
- Preset: themes/default.ts dan themes/dark.ts

nuxt-ui-lib

Library komponen UI enterprise untuk Nuxt 3

- EButton — 5 variant, 5 size, loading spinner, block prop
- EInput — v-model, label, error/hint, useId() a11y, SSR-safe
- ETable<T> — sortable, slot #cell-{key}, empty state, generic
- EModal — Teleport body, Transition, closeOnOverlay, 3 slot
- EBadge — 6 warna dengan color-mix() tint
- EPageContainer + tailwind.config.ts (CSS vars → Tailwind)

nuxt-composables

Kumpulan Vue 3 composable siap pakai untuk enterprise

- useForm<T> — Rules validator, errors reactive, handleSubmit async, reset, touched
- usePagination — page/size/total, totalPages computed, next/previous/goTo
- useDebounce<T> — delay configurable (default 300ms)
- usePermission — useState SSR-safe, hasRole, hasPermission, hasAll, isAdmin
- useLocalStorage<T> — SSR-safe, deep watch, JSON serialize/deserialize
- useThrottle<T> — interval configurable (default 300ms)

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Tech Stack

BACKEND FRAMEWORK

Quarkus 3.15.1

Java 21 · Eclipse Temurin

RESTEasy Reactive · Hibernate ORM Panache

FRONTEND FRAMEWORK

Nuxt 3 + Vue 3

TypeScript 5 · @nuxt/module-builder

SSR-safe · Auto-import composables

DATABASE

PostgreSQL 16

Flyway migrations · JSONB support

Hibernate ORM · JDBC health check

SECURITY

Multi-Layer Encryption

AES-256-GCM · RSA-2048 OAEP

BCrypt · SHA-256/512 · HMAC-SHA256

CONTAINER

Docker + Rancher Desktop

Multi-stage Dockerfile · docker-compose

WSL 2 · moby engine 29.1.4

BUILD TOOLS

Maven 3.9 + npm ws

GraalVM native profile · Makefile

npm workspaces · tsconfig paths

Fitur Enterprise yang Diimplementasikan

✓ Microservice Architecture

Setiap modul independen dengan pom.xml/package.json sendiri, bisa di-deploy terpisah via Docker. Tidak ada coupling antar modul.

✓ Container Friendly

Multi-stage Dockerfile di setiap modul. docker-compose.yml dengan healthcheck, environment variables, dan volume management.

✓ Binary Apps Conversion

GraalVM native profile (-Pnative) di Maven parent POM untuk kompilasi ke native binary tanpa JVM overhead.

✓ Database Adapter

BaseEntity + BaseRepository generik dengan paginasi, soft-delete support, audit trail otomatis, dan JDBC health probe.

✓ Encryption Adapter

AES-256-GCM (data at rest), RSA-2048 OAEP (key exchange), BCrypt (password), HMAC-SHA256 (integrity), constant-time compare.

✓ Themes Adapter

Design tokens (JSON) disimpan di DB → quarkus-theme-api → diambil nuxt-theme-lib → diterapkan sebagai CSS variables ke seluruh UI.

Arsitektur Database — quarkus-theme-api

Tabel / Kolom	Tipe	Keterangan
themes.id	BIGSERIAL PK	Primary key auto-generate
themes.name	VARCHAR(100)	Nama tampilan tema
themes.slug	VARCHAR(100) UNIQUE	Identifier URL-safe, unik
themes.active	BOOLEAN	Hanya satu tema aktif dalam satu waktu
themes.mode	VARCHAR(20)	light / dark
themes.tokens_json	JSONB	Design tokens: colors, typography, spacing, radius
themes.tenant_id	VARCHAR(100)	Multi-tenant isolation (nullable)
themes.created_at	TIMESTAMP	Diisi otomatis saat INSERT
themes.updated_at	TIMESTAMP	Diisi otomatis saat UPDATE

Struktur Setiap Modul

SOURCE CODE

src/main/java/ (Backend)

src/runtime/ (Frontend Nuxt module)

KONFIGURASI

.env + .env.example

application.properties / tsconfig.json

CONTAINER

Dockerfile (multi-stage)

docker-compose.yml per modul

BUILD

pom.xml / package.json

Maven parent inheritance · npm workspace

DOKUMENTASI

README.md lengkap

Instalasi, API, contoh kode, konfigurasi

TESTING

src/test/ (JUnit 5)

@QuarkusTest + REST Assured

Quick Start

```
# Clone & setup
git clone https://github.com/JackD-Creator/Java-Devx.git
cd Java-Devx/enterprise-lib

# Install frontend dependencies
npm install

# Build semua backend module
mvn clean install -DskipTests

# Jalankan quarkus-theme-api + PostgreSQL via Docker
cd backend-lib/quarkus-theme-api && docker compose up -d

# Test API
curl http://localhost:8080/api/themes/active
curl http://localhost:8080/q/health
```